

An abstract, high-contrast blue liquid background with swirling, organic patterns and bright highlights, resembling a close-up of a liquid surface or a microscopic view of a fluid.

RecipeRAG AI – Intelligent Document Q&A Recipe Generator

RAG-Powered Culinary Intelligence · Built on IBM's Enterprise AI Stack

The left side of the slide features a vertical panel with an abstract, high-contrast blue liquid background. The liquid appears to be flowing and swirling, creating a sense of movement and depth. The colors range from deep navy blue to bright, reflective highlights.

The Problem: Cooking is Broken for Modern Lifestyles

Overwhelm

Millions of recipes exist — yet people still don't know what to cook with what they have

Unaddressed Needs

Dietary restrictions, allergies, and nutrition goals go ignored by generic recipe apps

Disconnected Tools

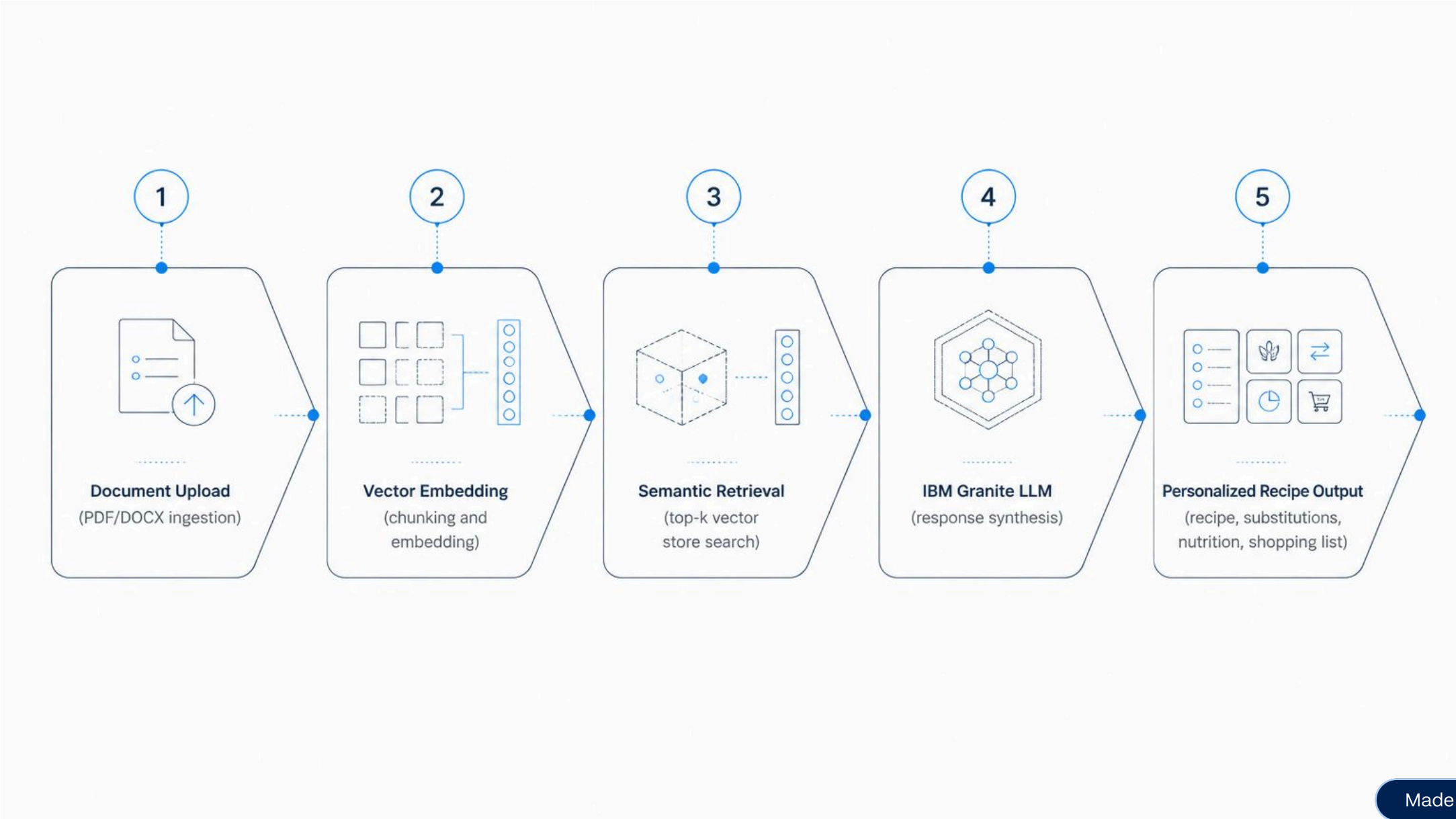
No single tool connects personal documents, preferences, and real-time Q&A



Our Solution: RAG-Powered Culinary Intelligence

- **Upload Any Recipe Document**
Ask anything → get personalized, context-aware answers instantly
- **Grounded in Your Documents**
Retrieval-Augmented Generation ensures every response is sourced from your actual files
- **Zero Hallucinations**
Real ingredients. Real recipes. Real nutrition — every time.

System Architecture



How It Works: 4-Step Workflow



1. Upload

Drop in recipe PDFs, cookbooks, or nutrition guides



2. Ask

"What can I make with chicken and no gluten?"



3. Retrieve

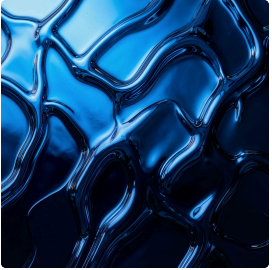
RAG fetches the most relevant document chunks instantly



4. Generate

IBM Granite crafts a tailored recipe with substitutions & nutrition

6 Powerful Features



Document Upload

PDF, DOCX, TXT recipe ingestion



Conversational Q&A

Multi-turn chat with memory



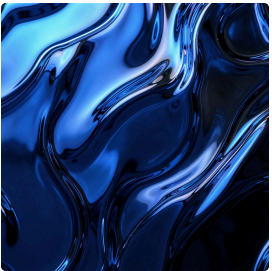
Personalized Recipes

Tailored to diet, allergies & preferences



Ingredient Substitutions

Smart swaps for what you have



Nutrition Analysis

Calories, macros, micronutrients per recipe



Shopping List Generator

Auto-generated, organized grocery lists

Powered by IBM's AI Stack



IBM Granite Models

Enterprise-grade LLM for accurate, safe, and grounded recipe generation



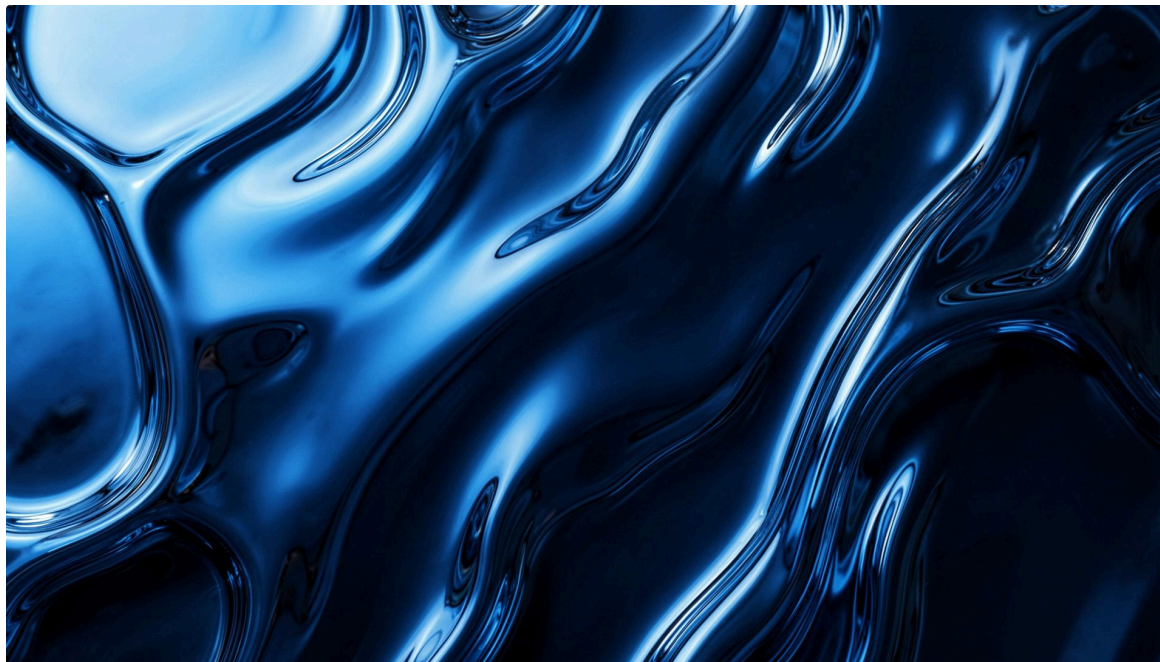
IBM Langflow

Visual RAG pipeline builder enabling rapid prototyping and deployment



IBM Orchestrate

Intelligent agent orchestration coordinating tasks across the full AI workflow



▶ LIVE DEMO SNAPSHOT

See It in Action

01

User Uploads

A 50-page vegan cookbook PDF ingested in seconds

02

User Asks

"High-protein dinner under 500 calories, no soy?"

03

RecipeRAG Delivers

Recipe + substitutions + macros + shopping list — **in under 3 seconds**

Future Scope: Where RecipeRAG Goes Next



Multi-Language Support

Regional cuisines delivered in native languages



Grocery API Integration

One-click order directly from your AI-generated shopping list



Image-Based Detection

Snap a photo of your fridge — get a recipe instantly



Community Recipe Graph

Crowdsourced RAG knowledge base for a living culinary intelligence



Clinical Nutrition Mode

Hospital & dietitian integrations for medical-grade guidance



The Future of Cooking is Intelligent — Build It With Us



Any Document

Transforms into a personal
culinary assistant



Enterprise-Ready

Built on IBM's scalable, safe,
production-grade AI stack



For Everyone

Making intelligent, personalized nutrition universally accessible